

Global Facilities - Design & Construction Standard - Chemical - Approved & Compatible Material Specifications for Chemical and Slurry Systems

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Owner(s)	ENGINEERING	Target Audience:	All Facilities
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[ECN History](#)[Submit a Change Request](#)

1.0 Purpose

This Standard Specification provides guidance on the materials and components used in Chemical and Slurry Systems. In addition to the materials and component advice, the document should provide guidance and information on the recommended vendors for these components. The goal is to improve and share knowledge as well as guide and advise for future projects.

This document is part of Micron's general specification standards developed to assist sites and suppliers to provide the best equipment reliability, as well as process and physical safety.

This document supplements the Global Design Standards for the individual system types as stipulated below:

1. Chemical Blender System
2. Chemical Distribution System
3. Bulk Chemical Unloading & Transfer System
4. Slurry Distribution System

These systems use many components in common and these systems have many shared chemistries and applications. Components that are specific to certain system types will not necessarily be referenced here.

Note: This specification is just a guide. All Local Standards must be adhered to. If any sites practice (or require) higher standards/ specifications/ regulations on compliance/etc., those sites may have to follow their existing standards/ specifications/ regulations on compliance/ etc. in order to meet site specific requirements, standard and specifications.

2.0 Scope

2.1 General Scope:

This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects. However, based on requirements and applicability with existing design, this document can also be used for Brownfield and other small expansion projects.

2.2 Technical Scope:

This document should be used only as a guide. Site personnel and suppliers are ultimately responsible for material compatibility in the systems they provide for the applications that they are provided. It provides technical guidance on materials based on past experience and Best-Known Practices and Methods.

Note: This document does not apply to systems for organic chemicals and deposition and other products that are supplied by dedicated cabinets and are normally associated with the gas systems

3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team

Term/Acronym	Definition / Description
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
APV	New PTFE APV (Advanced Performance Vessel), aka Modified PTFE
BKM	Best Known Method
ET	Etch: chemical etched sheet on the back of lining material for bonding to shell
EP	Electro Polished
FRP	Fiber - reinforced plastic
FEP	Fluorinated Ethylene Propylene
FCT	Facilities Central Team
FM	Factory Mutual
GN2	General Nitrogen
GB	Glass Backing, intermediary sheet bonded between tank liner and shell
GFPP	Glass - filled Polypropylene
GFTT	Global Facilities Technology Team
HMI	Human Machine Interface
HDPE	High Density Polyethylene
LPP	Low Permeation Properties, two - layer PFA tube
MHB	Main Header Box
MHEB	Main Header End Box
N2	Nitrogen
PLC	Programmable Logic Controller
PN2	Process Nitrogen
PTFE	Polytetrafluoroethylene
PLP	Purified Low Permeation, aka Modified PTFE
PTFM	Modified PTFE
PFA	Perfluoroalkoxy Alkane
PP	Polypropylene
PVC	Polyvinyl Chloride
PVDF	Polyvinylidene Difluoride
SPOF	Single Point of Failure
SS	Stainless Steel
TMAH	Tetramethyl Ammonium Hydroxide
UN2	Utility Nitrogen
VMB	Valve Manifold Box

4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead	• Develop and maintain the Standard documents. • Incorporate BKMs, lessons learned and emerging technologies into the standard. • Solicit, receive, and incorporate information from all other stakeholders on component and vendor quality. • Perform periodic review and update document.
FCT Lead	• Review and provide technical feedback on the Standard • Update relevant BKMs • Solicit information from site and forward on component and vendor quality and performance and any new components available.
Site Project Construction Team	• Review and provide technical feedback on the Standard • Incorporate Standard details into Project Engineering, Design, and Construction as appropriate • Document deviations, exclusions, and exceptions to the Standard • Share new lessons learned to GFTT and FCT Lead
Site Facilities Team	• Use Standard as technical reference for new and existing systems.

5.0 Approved and Compatible material specifications for Chemical & Slurry Systems.

S/N	COMPONENTS	ITEMS	ACIDS	BASES / OXIDIZER	CORROSIVE SOLVENTS	SOLVENTS	SLURRY
1	Tubing	Materials	PFA-450HP/451HP, AP231SH	PFA-450HP/451HP, AP231SH	PFA-450HP/451HP, AP231SH or Anti-static PFA	SS316L EP (10 Ra max) for ACE, FN-DP001, FN-RP002, PGMEA, RER900 / EBR300(CHN), LA-95	PFA-451HP or equivalent
		Suppliers	Entegris, Furon, GEMU, Nichias, Parker, Pureline, Nippon Pillar, or equivalent	Entegris, Furon, GEMU, Nichias, Parker, Pureline, Nippon Pillar, or equivalent	Entegris, Furon, GEMU, Nichias, Parker, Pureline, Nippon Pillar, or equivalent	Kuze / Dockweiler	Entegris or equiv.
		Notes/ Exceptions	PFA950/951, LPP For HF49%, HCl, HNO3		Anti-Static PFA for Acetic Acid, PK-HUZ, ACE002, GENET	IPA/PGMEA Depending on local code	
2	Cabinet / VMB / MHB / MHEB	Materials	PVC or PP with clear PVC panels	PVC or PP with clear PVC panels	SS-304L with reinforced static dissipative glass panels	SS-304L with reinforced static dissipative glass panels	PVC or PP with clear PVC panels
		Suppliers	Mitsubishi, Rochling, Simona or equiv.	Mitsubishi, Rochling, Simona or equiv.	various	various	Mitsubishi, Rochling, Simona or equiv.
		Notes/ Exceptions	FM4910 rated if installed in sub-fab	FM4910 rated if installed in sub-fab			FM4910 rated if installed in sub-fab
3	Secondary Containment	Materials	CLEAR PVC, sch 40 min.	CLEAR PVC, sch 40 min.	CLEAR PVC/SS304 dependent on chemical	N/A	CLEAR PVC, sch 40 min.
		Suppliers	George Fisher, Hershey Spears, Landmark Plastics or equivalent	George Fisher, Hershey Spears, Landmark Plastics or equivalent	George Fisher, Hershey Spears, Landmark Plastics or equivalent	Kuze /Valex /Dockweiler	George Fisher, Hershey Spears, Landmark Plastics or equivalent
		Notes/ Exceptions					

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S/N	COMPONENTS	ITEMS	ACIDS	BASES / OXIDIZER	CORROSIVE SOLVENTS	SOLVENTS	SLURRY
4	Wetted Fittings	Type	Super Pillar 300 /Primelock	Super Pillar 300 /Primelock	Super Pillar 300 /Primelock	VCR or welded	Super Pillar 300 /Primelock
		Suppliers	Nippon-Pillar, Entegris	Nippon-Pillar, Entegris	Nippon-Pillar, Entegris	various	Nippon-Pillar, Entegris
		Notes/ Exceptions	Flaretek fittings can only be used if above fittings options are not available. If flaretek are used, GEMU supplier can be used.	Flaretek fittings can only be used if above fittings options are not available. If flaretek are used, GEMU supplier can be used.	Flaretek fittings can only be used if above fittings options are not available. If flaretek are used, GEMU supplier can be used.		Flaretek fittings can only be used if above fittings options are not available. If flaretek are used, GEMU supplier can be used.
5	Chemical Process Valves	Type	PFA/PTFE	PFA/PTFE	PFA/PTFE	SS316-EP	PFA/PTFE
		Suppliers	Entegris (Integra Plus WS) GEMU (Weir type), CKD, Swagelok, or equivalent	Entegris (Integra Plus WS) GEMU (Weir type), CKD, Swagelok, or equivalent	Entegris (Integra Plus WS) GEMU (Weir type), CKD, Swagelok, or equivalent	SS valves (GEMU/FUJIKIN)	Entegris (Integra Plus WS) GEMU (Weir type), Swagelok or equivalent
		Notes/ Exceptions	GEMU valves shall be Seat not Weir type for pneumatic pump suction valve	GEMU valves shall be Seat not Weir type for pneumatic pump suction valve	GEMU valves shall be Seat not Weir type for pneumatic pump suction valve		GEMU valves shall be Seat not Weir type for pneumatic pump suction valve
6	Drain Connections / Manifolds	Materials	PP: Schedule 40 minimum	PP: Schedule 40 minimum	SS304/PP depending on chemistry	SS304	PP with clear PVC union sections spaced out periodically
		Suppliers	various	various	various	various	various
		Notes/ Exceptions	Use PVDF for H2SO4				

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S/N	COMPONENTS	ITEMS	ACIDS	BASES / OXIDIZER	CORROSIVE SOLVENTS	SOLVENTS	SLURRY
7	Pneumatic Tubing (wetted area)	Materials	PFA/FEP	PFA/FEP	PFA/FEP	PFA/FEP	PFA/FEP
		Suppliers	Various	Various	Various	Various	Various
		Notes/ Exceptions					
8	Pneumatic Transfer /Dispense Pumps	Materials	Teflon PFA/PTFE	Teflon PFA/PTFE	Teflon PFA/PTFE	Teflon PFA/PTFE or SS	Teflon PFA/PTFE
		Suppliers	Trebor /Nippon- Pillar or equivalent	Trebor /Nippon- Pillar or equivalent	Trebor /Nippon- Pillar or equivalent	Trebor /Nippon- Pillar or equivalent	ARO/ Verder/ Nippon-Pillar or equivalent
		Notes/ Exceptions	No rubber /EPDM backing for diaphragm for high purity application	No rubber /EPDM backing for diaphragm for high purity application	No rubber /EPDM backing for diaphragm for high purity application	No rubber /EPDM backing for diaphragm for high purity application	PFA with Santoprene backed diaphragm for Transfer pumps
9	Centrifugal Pumps	Materials	PTFE/PFA/ECTFE coating	PTFE/PFA/ECTFE coating	PTFE/PFA/ECTFE coating	PTFE/PFA/ECTFE coating	PTFE/PFA/ECTFE coating
		Suppliers	Levitronix	Levitronix	Levitronix	Levitronix	Levitronix
		Notes/ Exceptions					
10	Drain/Sump pumps	Materials	various plastic (PP/PFA/PTFE)	various plastic (PP/PFA/PTFE)	various plastic (PP/PFA/PTFE)	SS	various plastic (PP/PFA/PTFE)
		Suppliers	ARO, Verder or equivalent	ARO, Verder or equivalent	ARO, Verder or equivalent	ARO, Verder or equivalent	ARO, Verder or equivalent
		Notes/ Exceptions	Santoprene or Santoprene backed diaphragm	Santoprene or Santoprene backed diaphragm	Santoprene or Santoprene backed diaphragm	SS body	Santoprene or Santoprene backed diaphragm

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11	Filter Housings /Filter housing Lock nut, filter housing O-ring	Materials	PFA	PFA	PFA	PFA/SS316L EP	PFA
		Suppliers	Pall, Entegris	Pall, Entegris	Pall, Entegris	Pall, Entegris	Pall, Entegris
		Notes/ Exceptions	PVDF Nut	PVDF nut for most; GFPP/PTFE nut for NH4OH & TMAH	PVDF Nut	Depending on Fire Codes and low conductive	PVDF nut
12	Solenoid Valves	Materials	-	-	-	-	-
		Suppliers	Festo, SMC or equivalent	Festo, SMC or equivalent	Festo, SMC or equivalent	Festo, SMC or equivalent	Festo, SMC or equivalent
		Notes/ Exceptions					
13	Pressure Transducers	Materials	PFA/PTFE	PFA/PTFE	PFA/PTFE	PFA/PTFE	PFA/PTFE
		Suppliers	GEMU /Entegris /Surpass	GEMU /Entegris /Surpass	GEMU /Entegris /Surpass	GEMU /Entegris /Surpass	GEMU /Entegris /Surpass
		Notes/ Exceptions	Sapphire interface for HF49% application (Entegris)				
14	Chem Flowmeters (Contact)	Materials	PFA/PTFE/CTFE	PFA/PTFE/CTFE	PFA/FEP	PFA/FEP	PFA/PTFE/CTFE
		Suppliers	Entegris /Levitronix	Entegris /Levitronix	Entegris /Levitronix	Entegris /Levitronix	Entegris/Levitronix
		Notes/ Exceptions					
15	Chem Flowmeters (Non-contact)	Materials	-	-	-	-	-
		Suppliers	Flexim /Levitronix /Keyence or equivalent	Flexim /Levitronix /Keyence or equivalent	Flexim /Levitronix /Keyence or equivalent	Flexim /Levitronix /Keyence or equivalent	Flexim /Levitronix /Keyence or equivalent
		Notes/ Exceptions			Intrinsically safe	Intrinsically safe	
16	Back Pressure Regulators (Chemical)	Materials	PFA/PTFE	PFA/PTFE	PFA/PTFE	PFA/PTFE	PFA/PTFE
		Suppliers	Parker/equivalent	Parker/equivalent	Parker/equivalent	Parker/equivalent	Parker/equivalent
		Notes/ Exceptions					

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S/N	COMPONENTS	ITEMS	ACIDS	BASES / OXIDIZER	CORROSIVE SOLVENTS	SOLVENTS	SLURRY
17	Drum Stirrer Shafts	Materials	N/A	N/A	N/A	N/A	SS304/plastic
		Suppliers	N/A	N/A	N/A	N/A	various/slurry supplier recommendation
		Notes/ Exceptions					
18	Tank Shell	Materials	SS304 with Epoxy or Polyurethane chemical resistant coating or equivalent	SS304 with Epoxy or Polyurethane chemical resistant coating or equivalent	SS304 with Epoxy or Polyurethane chemical resistant coating or equivalent	SS316L (EP)	HDPE/PP-natural or equivalent
	Tank Liner	Materials	PLP-ET; APV-GB; PTFM-GB or PFA- GB; 2.4mm minimum thickness' 3.0 mm thickness for HCl/HF/HNO3	PLP-ET; APV-GB; PTFM-GB or PFA- GB; 2.4mm minimum thickness	PLP-ET; APV-GB; PTFM-GB or PFA- GB; 2.4mm minimum thickness	N/A	N/A if HDPE/PP.
	Tanks	Suppliers	Allied Supreme, Edlon, Entegris, NGC, Sunfluoro, Valqua, Fluorotech, Yodagawa or equivalent	Allied Supreme, Edlon, Entegris, NGC, Sunfluoro, Valqua, Fluorotech, Yodagawa or equivalent	Allied Supreme, Edlon, Entegris, NGC, Sunfluoro, Valqua, Fluorotech, Yodagawa or equivalent	SFI (USA), Allied Supreme, Edlon, NGC, Sunfluoro, Valqua, Fluorotech, or equivalent	Various
	Tanks	Notes/ Exceptions	For HCL: Entegris FluoroPure Tanks with PFA loose liner 3mm thick new and HDPE shell approved; All nozzles welded instead of flanged; Exhausted enclosure	All other new tank manufacturers not listed above shall be qualified/approved by Micron.	All other new tank manufacturers not listed above shall be qualified/approved by Micron.	All other new tank manufacturers not listed above shall be qualified/approved by Micron.	

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S/N	COMPONENTS	ITEMS	ACIDS	BASES / OXIDIZER	CORROSIVE SOLVENTS	SOLVENTS	SLURRY
19	Tank Level Sensors (Ultra Sonic/Radar)	Materials	PTFE interface	PTFE interface	PTFE interface	SS	N/A
		Suppliers	Vega / Endress+Hauser or equivalent	Vega / Endress+Hauser or equivalent	Vega / Endress+Hauser or equivalent	Vega / Endress+Hauser or equivalent	Vega / Endress+Hauser or equivalent
		Notes/ Exceptions					
20	Tank Level Sensors (Discrete non- contact)	Materials	N/A	N/A	N/A	N/A	N/A
		Suppliers	IFM	IFM	IFM	IFM	IFM
		Notes/ Exceptions			Intrinsically safe	Intrinsically safe	
21	Mechanical Float Leak sensors	Materials	PP/PFA/PTFE	PTFE	PP/PFA/PTFE	SS	PP/PFA/PTFE
		Suppliers	FineTek, Kansai Automation	FineTek, Kansai Automation	FineTek, Kansai Automation	FineTek, Kansai Automation	FineTek, Kansai Automation
		Notes/ Exceptions			Intrinsically safe	Intrinsically safe	
22	Door Sensors	Materials	Magnetic type	Magnetic type	Magnetic type	Magnetic type	Magnetic type
		Suppliers	IFM & various	IFM & various	IFM & various	IFM & various	IFM & various
		Notes/ Exceptions			Intrinsically safe	Intrinsically safe	
23	Programmable Logic Controllers	Materials	N/A	N/A	N/A	N/A	N/A
		Suppliers	Allen Bradley, GE Fanuc, Mitsubishi, Omron, Siemens or equivalent	Allen Bradley, GE Fanuc, Mitsubishi, Omron, Siemens or equivalent	Allen Bradley, GE Fanuc, Mitsubishi, Omron, Siemens or equivalent	Allen Bradley, GE Fanuc, Mitsubishi, Omron, Siemens or equivalent	Allen Bradley, GE Fanuc, Mitsubishi, Omron, Siemens or equivalent
		Notes/ Exceptions	Siemens is the first choice	Siemens is the first choice	Siemens is the first choice	Siemens is the first choice	Siemens is the first choice
24	Load Cells/Scale Platforms	Materials	-	-	-	-	-
		Suppliers	Hardy, Mettler Toledo, or equivalent	Hardy, Mettler Toledo, or equivalent	Hardy, Mettler Toledo, or equivalent	Hardy, Mettler Toledo, or equivalent	Hardy, Mettler Toledo, or equivalent
		Notes/ Exceptions					

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S/N	COMPONENTS	ITEMS	ACIDS	BASES / OXIDIZER	CORROSIVE SOLVENTS	SOLVENTS	SLURRY
25	Pressure Regulators (Process N2)	Materials	SS or plastic	SS or plastic	SS or plastic	SS or plastic	SS or plastic
		Suppliers	Parker, SMC or equivalent	Parker, SMC or equivalent	Parker, SMC or equivalent	Parker, SMC or equivalent	Parker, SMC or equivalent
		Notes/ Exceptions					
26	Conductivity Probes	Materials	PFA	PFA			PFA
		Suppliers	Horiba / Endress+Hauser / ABB	Horiba / Endress+Hauser / ABB	N/A	N/A	Horiba / Endress+Hauser / ABB
		Notes/ Exceptions					
27	Refractometers (Concentration meters)	Materials	PFA/PTFE/sapphire	PFA/PTFE/sapphire			PFA/PTFE/sapphire
		Suppliers	K-patents, KxS or equivalent	K-patents, KxS or equivalent	N/A	N/A	K-patents, KxS or equivalent
		Notes/ Exceptions					
28	Titration	Suppliers	N/A	N/A	N/A	N/A	Entegris (SemiChem)
29	Temperature Probes/RTD's (wetted)	Materials	Stainless steel probe with PFA lining	Stainless steel probe with PFA lining	Stainless steel probe with PFA lining	Stainless steel probe with PFA lining	Stainless steel probe with PFA lining
		Suppliers	EI Sensor Technologies	EI Sensor Technologies	EI Sensor Technologies	EI Sensor Technologies	EI Sensor Technologies
		Notes/ Exceptions			Intrinsically safe	Intrinsically safe	

6.0 Document Control

6.1 Approval

GLOBAL_FAC_GFTT_APPR

6.2 Notification

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

6.3 Final Viewer Access

All Micron team members in all sites

6.4 Retention

Adherence to the requirements specified in
[Global Facilities - Document Control Procedures](#)

6.5 Review

Biennially (two years)

7.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	08/22/22